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COUPLANT COATING MECHANISM AND ULTRASONIC IMAGING SYSTEM

Abstract

A couplant coating mechanism including an engagement portion engaging with a scanning probe, the engagement portion being provided with a first channel; and an accommodation portion connected to the engagement portion, the accommodation portion including a main body portion for accommodating a couplant, an internal space of the main body portion being in communication with the first channel, and the couplant accommodated in the main body portion flowing out along the first channel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claim priority to Chinese Patent Application No. 202410244489.0, which was file on Mar. 4, 2024 at the Chinese Patent Office. The entire contents of the above-listed application are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present application relates to the field of medical imaging, and in particular to a couplant coating mechanism and an ultrasonic imaging system.

BACKGROUND

[0003] Ultrasonic imaging is one of the important means for imaging the interior of the body of a person to be scanned. Generally, ultrasonic imaging systems use ultrasonic transducers to convert electrical energy into ultrasonic pulses. The ultrasonic pulses are sent to the interior of the body of the person to be scanned, and then echo signals are generated. The echo signals are received by a transducer element and then converted into electrical signals. The electrical signals are processed by a specialized processing device to then form a desired ultrasonic image.

[0004] Ultrasonic imaging systems have important applications in scanning many organs of the body. For example, a full-field breast ultrasonic scanning device may be used to image breast tissue in one or more planes. During a full-field breast ultrasonic scanning process, it is generally necessary to apply a couplant to the body of a patient prior to scanning.

[0005] It should be noted that the above introduction of the background is only for the convenience of clearly and completely describing the technical solutions of the present application, and for the convenience of understanding for those skilled in the art. The above technical solutions are not considered to be well known to those skilled in the art merely because they are set forth in the Background of the present application.

SUMMARY

[0006] The inventors have found that, at present, a method for applying a couplant to a patient comprises a technician squeezing the couplant from a bottle to apply the couplant to the body of the patient, and then using a tool to apply the couplant slowly and uniformly to the body of the patient. This process occupies the total time of scanning and reduces scanning efficiency, and there is a problem that uneven application caused by manual operation affects the scanning effect.

[0007] In view of at least one of the above problems or other similar problems, embodiments of the present application provide a couplant coating mechanism and an ultrasonic imaging system. Automatic application of a couplant is implemented by adding the couplant coating mechanism to a scanning probe, thereby saving scanning time, avoiding couplant waste, and further improving scanning efficiency and user experience.

[0008] According to one aspect of the embodiments of the present application, provided is a couplant coating mechanism. The couplant coating mechanism comprises:

[0009] An engagement portion engaging with a scanning probe, the engagement portion being provided with a first channel; and

[0010] An accommodation portion connected to the engagement portion, the accommodation portion comprising a main body portion for accommodating a couplant, an internal space of the main body portion being in communication with the first channel, and the couplant accommodated in the main body portion flowing out along the first channel.

[0011] In some embodiments, the accommodation portion further comprises an extension portion connected to the main body portion, the extension portion is provided with a second channel that is in communication with the internal space of the main body portion, and the internal space of the main body portion is in communication with the first channel via the second channel.

[0012] In some embodiments, there are two extension portions oppositely disposed, and each extension portion is provided with the second channel.

[0013] In some embodiments, the engagement portion comprises two wall portions, and a connection portion connecting the two wall portions; and a gap between the two wall portions is used for accommodating the scanning probe, and each wall portion is provided with the first channel.

[0014] In some embodiments, there are a plurality of first channels disposed extending from top to bottom.

[0015] In some embodiments, extending ends of the two wall portions each have a couplant outlet in communication with the first channel.

[0016] In some embodiments, the couplant outlet is an integral structure formed along the extending end, and the curvature of the couplant outlet matches that of an extending end of the scanning probe.

[0017] In some embodiments, the height of the couplant outlet is higher than the height of an extending end of the scanning probe.

[0018] In some embodiments, the accommodation portion further comprises:

[0019] An extruding portion, the extruding portion being disposed inside the main body portion and capable of moving to extrude the couplant; and

[0020] A driving portion, the driving portion driving the extruding portion to move.

[0021] In some embodiments, the pressing portion comprises a first flipping plate and a second flipping plate that divide the internal space of the main body portion into two parts, and the driving portion comprises a first motor and a second motor; and the first motor is capable of driving the first flipping plate to move, so that the couplant on a side of the first flipping plate flows out, and the second motor is capable of driving the second flipping plate to move, so that the couplant on a side of the second flipping plate flows out.

[0022] In some embodiments, the main body portion is provided with two openings, the two openings being in communication with two internal spaces, respectively, which are separated by the first flipping plate and the second flipping plate.

[0023] In some embodiments, the main body portion further comprises:

[0024] An electrical connection portion, the electrical connection portion being connected to the driving portion to transmit electrical energy to the driving portion,

[0025] The electrical connection portion comprising at least one of:

[0026] An electrical receiving terminal, the electrical receiving terminal being disposed on an inner side of the main body portion or the engagement portion and cooperating with a power supply terminal of the scanning probe to receive the electrical energy; and

[0027] A cable, the cable receiving electrical energy from outside the couplant coating mechanism.

[0028] According to another aspect of the embodiments of the present application, provided is an ultrasonic imaging system. The system comprises:

[0029] A scanning assembly, comprising a scanning probe; and

[0030] The couplant coating mechanism according to any one of the foregoing embodiments.

[0031] In some embodiments, the scanning assembly further comprises a frame, the frame accommodating the scanning probe and a driving device, the driving device driving the scanning probe to move within the frame to perform ultrasonic scanning.

[0032] One advantageous effect of the embodiments of the present application is that automatic application of a couplant is implemented by adding the couplant coating mechanism to the scanning probe, thereby saving scanning time, avoiding couplant waste, and further improving scanning efficiency and user experience.

[0033] With reference to the following description and drawings, specific implementations of the present application are disclosed in detail. It should be understood that the implementations of the present application are not limited in scope thereby. Within the scope of the spirit and clauses of the

appended claims, the embodiments of the present application include many changes, modifications, and equivalents.

[0034] The features described and/or illustrated for one implementation may be used in one or more other implementations in the same or similar manner, be combined with features in other embodiments, or replace features in other implementations.

[0035] It should be emphasized that the term “include/comprise/have”, when used herein, refers to the presence of features, integrated components, or assemblies, but does not preclude the presence or addition of one or more other features, integrated components, or assemblies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0036] The above and other objects, features and advantages of the embodiments of the present application will become more apparent from the following detailed description with reference to the drawings, in which:

[0037] FIG. 1 is a perspective view of an ultrasonic imaging system according to embodiments of the present application;

[0038] FIG. 2 is a block diagram of the ultrasonic imaging system according to embodiments of the present application;

[0039] FIG. 3 is a schematic diagram of a couplant coating mechanism according to embodiments of the present application;

[0040] FIG. 4 is a schematic diagram of an installation position of a couplant coating mechanism according to embodiments of the present application;

[0041] FIG. 5 is a schematic diagram of a couplant coating mechanism viewed from another angle;

[0042] FIG. 6 is a schematic exploded view of a couplant coating mechanism according to embodiments of the present application;

[0043] FIG. 7 is a schematic diagram of a partial structure of an accommodation portion of a couplant coating mechanism according to embodiments of the present application;

[0044] FIG. 8 is another schematic diagram of a couplant coating mechanism according to embodiments of the present application; and

[0045] FIG. 9 is still another schematic diagram of a couplant coating mechanism according to embodiments of the present application.

DETAILED DESCRIPTION

[0046] The foregoing and other features of the embodiments of the present application will become apparent from the following description with reference to the drawings. In the description and drawings, specific implementations of the present application are disclosed in detail, and part of the implementations in which the principles of the present application may be employed are indicated. It should be understood that the present application is not limited to the described implementations, and include all modifications, variations, and equivalents which fall within the scope of the appended claims.

[0047] In the embodiments of the present application, the terms “first”, “second”, “upper”, “lower”, etc. are used to distinguish different elements with respect to naming, but do not represent a spatial arrangement, a temporal order, etc. of these elements, and these elements should not be limited by these terms. The term “and/or” includes any and all combinations of one or more associated listed terms. The terms “comprise”, “include”, “have”, etc., refer to the presence of described features, elements, components, or assemblies, but do not exclude the presence or addition of one or more other features, elements, components, or assemblies.

[0048] In the embodiments of the present application, the singular forms “a” and “the”, etc., include plural forms, and should be broadly construed as “a type of” or “a class of” rather than

being limited to the meaning of “one”. Furthermore, the term “the” should be construed as including both the singular and plural forms, unless otherwise specified in the context. In addition, the term “according to” should be construed as “at least in part according to . . .” and the term “on the basis of” should be construed as “at least in part on the basis of . . .”, unless otherwise specified in the context.

[0049] The features described and/or illustrated for one embodiment may be used in one or more other embodiments in an identical or similar manner, combined with features in other embodiments, or replace features in other embodiments.

[0050] FIG. 1 shows a perspective view of an ultrasonic imaging system **102** according to some embodiments. As shown in FIG. 1, the body of the ultrasonic imaging system **102** may include a main device, a display **110**, an adjustable arm **106**, and a scanning assembly **108**. The main device may include a body frame **104**, an ultrasonic processor housing **105**, and an ultrasonic processor inside the housing **105**. The specific structure of each component will be illustrated in detail below.

[0051] The body frame **104**, the ultrasonic processor housing **105** containing the ultrasonic processor, a movable and adjustable support arm (for example, an adjustable arm) **106** including a hinge joint **114**, the scanning assembly **108** connected to a first end **120** of the adjustable arm **106** by means of a ball and socket connector (for example, a ball joint) **112**, and the display **110** connected to the body frame **104**. The display **110** is connected to the body frame **104** at a joining point where the adjustable arm **106** enters the body frame **104**. Since the display **110** is directly connected to the body frame **104** rather than the adjustable arm **106**, the display **110** does not affect the weight of the adjustable arm **106** and a balancing mechanism of the adjustable arm **106**. In one example, the display **110** is rotatable in horizontal and transverse directions (for example, rotatable around a central axis of the body frame **104**), but is not vertically movable. In an alternative example, the display **110** may also be vertically movable. Although FIG. 1 depicts the display **110** connected to the body frame **104**, in other examples, the display **110** may be connected to different components of the imaging system **102**, such as, connected to the ultrasonic processor housing **105**, or positioned away from the imaging system **102**.

[0052] In one embodiment, the adjustable arm **106** is configured and adapted such that the pressing/scanning assembly **108** (i) is neutrally buoyant in space, or (ii) has a light net downward weight (for example, 1-2 kg) for pressing the breast, while allowing easy user operation. In an alternative embodiment, the adjustable arm **106** is configured such that the scanning assembly **108** is neutrally buoyant in space during positioning of a scanner on tissue of a patient. Then, after the scanning assembly **108** is positioned, internal components of the imaging system **102** may be adjusted to apply a desired downward weight for pressing the breast and improving image quality. In one example, the downward weight (for example, a force) may be in the range of 2-11 kg.

[0053] As described above, the adjustable arm **106** includes the hinge joint **114**. The hinge joint **114** divides the adjustable arm **106** into a first arm portion and a second arm portion. The first arm portion is connected to the scanning assembly **108** and the second arm portion is connected to the body frame **104**. The hinge joint **114** allows the second arm portion to rotate relative to the second arm portion and the body frame **104**. For example, the hinge joint **114** allows the scanning assembly **108** to translate transversely and horizontally, but not vertically, relative to the second arm portion and the body frame **104**. In such manner, the scanning assembly **108** can rotate toward the body frame **104** or away from the body frame **104**. However, the hinge joint **114** is configured to allow the entire adjustable arm **106** (for example, the first arm portion and the second arm portion) to move vertically together as a whole (for example, translating upward and downward along with the body frame **104**).

[0054] The scanning assembly **108** may include a film assembly **118** having a film that is in a substantially tensioned state to be at least partially attached, for pressing the breast. The film assembly **118** has a bottom surface for contacting the breast, and when the bottom surface is in contact with the breast, the transducer sweeps over a top surface of the film to scan the breast. In

one example, the film is a tensioned fabric sheet.

[0055] The film assembly **118** may further include an outer frame and a film. The film is fixedly disposed in the outer frame, and the outer frame is detachably connected to the scanning assembly. In an ultrasonic imaging process performed by the ultrasonic imaging system, one side surface of the film can be at least partially in contact with an ultrasonic transducer, and another side surface of the film is at least partially in contact with a tissue to be scanned. Such an arrangement can ensure that the ultrasonic transducer transmits and receives signals with less attenuation, and can fix the breast to be scanned to facilitate scanning.

[0056] Optionally, the adjustable arm may include a potentiometer (not shown) to allow position and direction sensing performed by the pressing/scanning assembly **108**, or may use other types of position and direction sensing (such as gyroscope, magnetic, optical, and radio frequency (RF)). A fully functional ultrasonic engine may be provided within the ultrasonic processor housing **105**, and is configured to drive the ultrasonic transducer, and generate volumetric breast ultrasound data from a scan in conjunction with related position and orientation information. In some examples, volumetric scan data may be transmitted to another computer system by using any of a variety of data transmission methods known in the art so as to be further processed, or the volumetric scan data may be processed by the ultrasonic engine. A general-purpose computer/processor integrated with the ultrasonic engine may further be provided for general user interface and system control. The general-purpose computer may be a self-contained stand-alone unit, or may be remotely controlled, configured, and/or monitored by remote stations connected across networks.

[0057] FIG. **2** is a block diagram **200** that schematically illustrates various system components of the ultrasonic imaging system **102**. As shown in FIG. **2**, the ultrasonic imaging system **102** includes a scanning assembly **108**, a display **110**, and a scanning processor **210**. In one example, the scanning processor **210** may be included within an ultrasonic processor housing **105** of the imaging system **102**. As shown in FIG. **2**, the scanning assembly **108**, the display **110**, and the scanning processor **210** are separate components communicating with each other. However, in some embodiments, one or more of these components may be integrated (for example, the display and the scanning processor may be included in a single component).

[0058] In the example of FIG. **2**, the scanning assembly **108** includes at least a transducer module **220** and a driving device **240**. The transducer module **220** includes a transducer array of transducer elements, such as a piezoelectric element, which converts electrical energy into ultrasonic waves and then detects reflected ultrasonic waves.

[0059] The scanning assembly **108** may communicate with the scanning processor **210** to send raw scan data to an image processor. The scanning assembly **108** may optionally communicate with the display **110** so as to indicate a user to reposition the scanning assembly as described above, or to receive information from the user (via user input **244**).

[0060] In the example of FIG. **2**, the scanning processor **210** includes an image processor **212**, a memory **214**, display output **216**, and an ultrasonic engine **218**. The ultrasonic engine **218** may drive activation of the transducer elements of the transducer module **220**, and in some embodiments, the driving device **240** may be activated. Furthermore, the ultrasonic engine **218** may receive raw image data (for example, ultrasonic echoes) from the scanning assembly **108**. The raw image data may be sent to the image processor **212** and/or a remote processor (for example, via a network) and be processed to form a displayable image of a tissue sample. It should be understood that in some embodiments, the image processor **212** may be included in the ultrasonic engine **218**.

[0061] In the example of FIG. **2**, information may be transmitted from the ultrasonic engine **218** and/or the image processor **212** to the user of the imaging system **102** via the display output **216** of the scanning processor **210**. In one example, the user of the ultrasonic imaging system may include an ultrasonic technician, a nurse, or a physician such as a radiologist. For example, a processed image of scanned tissue may be sent to the display **110** via the display output **216**. In another

example, information (such as the progress of scanning) related to parameters of the scanning may be sent to the display **110** via the display output **216**. The display **110** may include a user interface **242** configured to display images or other information to the user. Furthermore, the user interface **242** may be configured to receive an input from the user (such as by means of a user input unit **244**), and send the input to the scanning processor **210**. In one example, the user input unit **244** may be a touch screen of the display **110**. However, other types of user input mechanisms are also possible, such as a mouse, a keyboard, and the like.

[0062] The scanning processor **210** may further include the memory **214**. The memory **214** may include movable and/or permanent devices, and may include an optical memory, a semiconductor memory, and/or a magnetic memory, etc. The memory **214** may include a volatile, non-volatile, dynamic, static, read/write, read only, random access, sequential access, and/or annex memory. The memory **214** may store non-transitory instructions executable by a controller or processor (such as a controller or the image processor **212**) so as to perform one or more methods or routines. The memory **214** may store raw image data received from the scanning assembly **108**, processed image data received from the image processor **212** or the remote processor, and/or additional information.

[0063] The embodiments of the present application provide a couplant coating mechanism.

[0064] FIG. **3** is a schematic diagram of a couplant coating mechanism according to the embodiments of the present application, FIG. **4** is a schematic diagram of an installation position of the couplant coating mechanism according to the embodiments of the present application, and FIG. **5** is a schematic diagram of the couplant coating mechanism viewed from another angle.

[0065] As shown in FIG. **3** to FIG. **5**, the couplant coating mechanism **400** according to the embodiments of the present application includes an engagement portion **410** engaging with a scanning probe P and an accommodation portion **420** connected to the engagement portion **410**. The engagement portion **410** is provided with a first channel **411**, the accommodation portion **420** includes a main body portion **421** for accommodating a couplant, an internal space of the main body portion **421** is in communication with the first channel **411**, and the couplant accommodated in the main body portion **421** flows out along the first channel **411**.

[0066] According to the foregoing embodiment, automatic application of a couplant is implemented by adding the couplant coating mechanism to the scanning probe, thereby saving scanning time, avoiding couplant waste, and further improving scanning efficiency and user experience.

[0067] In the foregoing embodiment, the engagement between the couplant coating mechanism and the scanning probe is not limited in manner, and may be implemented by means of an interference fit, or may be implemented by means of providing an engaging member, and any existing engagement manner is applicable to the above structure of the present application.

[0068] According to the above structure, an operator only needs to install the couplant coating mechanism on the scanning probe to implement automatic couplant coating, thereby improving the convenience of coating. In addition, since the couplant coating mechanism is engaged with the scanning probe, the couplant coating mechanism can be easily removed from the scanning probe when the couplant coating mechanism is not required, thereby improving convenience of use.

[0069] In some embodiments, as shown in FIG. **3**, the scanning probe P is a part of a scanning assembly **108**. The scanning assembly **108** includes a housing **310**, a transducer module **220**, and a module receiver **230**. The transducer module **220** and the module receiver **230** constitute the foregoing scanning probe P.

[0070] The housing **310** includes a frame **322** and a handle portion **324**, and the handle portion includes two handles **312**. The two handles **312** oppose each other across a transverse axis of the scanning assembly **108**, and the transverse axis is centered on the adjustable arm **106** and defined relative to a transverse axis **308**. The frame **322** is rectangular, and an inner periphery of the frame **322** defines an opening **314**. The opening **314** provides space (e.g., a void volume) for translating the module receiver **230** and the transducer module **220** during a scanning process. In another

example, the frame **322** can have another shape, such as a square having the square opening **314**. In addition, the frame **322** has a thickness defined between an inner periphery and an outer periphery of the frame **322**.

[0071] The frame **322** includes four sets of side walls (e.g., a set including inner and outer side walls, the inner side walls defining the opening **314**). In particular, the frame **322** includes a front side wall **326** and a rear side wall **328**, the rear side wall **328** is directly connected to the handle portion **324** of the housing **310**, and the front side wall **326** is opposite to the rear side wall **328** with respect to the horizontal axis **306**. The frame **322** further includes right and left side walls, the corresponding side walls opposing each other and both being in a plane defined by the vertical axis **304** and the transverse axis **308**.

[0072] The frame **322** of the housing **310** further includes a top side and a bottom side, and the top side and the bottom side are defined relative to the vertical axis **304**. The top side faces the adjustable arm **106**. The film **118** is disposed across the opening **314**. More specifically, the film **118** is connected to the bottom side of the frame **322**. In an example, the film **118** is a diaphragm that remains tensioned across the opening **314**. The film **118** may be made from a flexible but non-stretchable material, and the material is thin, waterproof, durable, highly acoustically transparent, resistant to chemical corrosion, and/or biocompatible. As described above, the bottom surface of the film **118** may contact a tissue (e.g., a breast) during scanning, and the upper surface of the film **118** may at least partially contact the transducer module **220** during scanning. As shown in FIG. 3, the film **118** is permanently connected to a hard-housing holding portion **119** surrounding the periphery of the film **118**. The holding portion **119** is connected to the bottom side of the frame **322**. In one example, the holding portion **119** can be fastened to a lip-like edge on the bottom side of the frame **322** of the housing **310**, so that the film **118** does not become unconnected during scanning, but is still removably connected to the frame **322**. The film **118** may not be permanently connected to the hard-housing holding portion **119**, and thus the film **118** may be connected to the frame **322** without the hard-housing holding portion **119**. Instead, the film **118** may be directly and removably connected to the frame **322**.

[0073] The handle portion **324** of the housing **310** includes the two handles **312** for moving the scanning assembly **108** in space and positioning the scanning assembly **108** on a tissue (e.g., on the body of a patient). In an alternative embodiment, the housing **310** may not include the handle **312**. In an example, the handle **312** may be integrally formed with the frame **322** of the housing **310**. In another example, the handle **312** and the frame **322** may be formed separately and then mechanically connected together to form the entire housing **310** of the scanning assembly **108**.

[0074] In the example of FIG. 3, the scanning assembly **108** may be connected to the adjustable arm **106** by means of a ball joint **112** (e.g., a ball and socket connector). Specifically, a top dome portion of the handle portion **324** is connected to the ball joint **112**. The top of the handle portion **324** includes a depression forming a socket, and a ball of the ball joint **112** is fit in the socket. The ball joint **112** is movable in multiple directions. For example, the ball joint **112** provides rotational motion of the scanning assembly relative to the adjustable arm **106**. The ball joint **112** includes a locking mechanism for locking the ball joint **112** in place, thereby holding the scanning assembly **108** stationary relative to the adjustable arm **106**. Furthermore, the ball joint **112** may also be configured to only rotate but not to move in multiple directions, such as oscillating.

[0075] Additionally, as shown in FIG. 3, the handle **312** of the handle portion **324** includes buttons for controlling scanning and adjusting the scanning assembly **108**. Specifically, a first handle of the handles **312** includes a first weight adjustment button **316** and a second weight adjustment button **318**. The first weight adjustment button **316** may reduce a load applied to the scanning assembly **108** from the adjustable arm **106**. The second weight adjustment button **318** may increase a load applied to the scanning assembly **108** from the adjustable arm **106**. Increasing the load applied to the scanning assembly **108** may increase the pressure and the amount of pressing applied to the tissue on which the scanning assembly **108** is placed. Furthermore, increasing the load applied to

the scanning assembly increases the effective weight of the scanning assembly on the tissue to be scanned. In one example, increasing the load may press a tissue of a patient, such as a breast. In such manner, varying amounts of pressure (e.g., load) may be applied consistently with the scanning assembly **108** during scanning, so as to obtain high quality images by using the transducer module **220**.

[0076] Before the scanning process, a user (e.g., an ultrasonic technician or physician) may position the scanning assembly **108** on a patient or a tissue. Once the scanning assembly **108** is properly positioned, the user may adjust a weight (e.g., adjusting an amount of pressing) of the scanning assembly **108** on the patient by using the first weight adjustment button **316** and/or the second weight adjustment button **318**. Then, the user may initiate the scanning process by means of additional control on the handle portion **324** of the housing **310**. For example, as shown in FIG. 3, the second handle of the handles **312** includes two additional buttons **330** (not separately shown). The two additional buttons **330** may include a first button for initiating a scan (e.g., once the scanning assembly has been placed on the tissue/patient and an amount of pressing has been selected) and a second button for stopping the scan. In one example, once the first button is selected, the ball joint **112** may be locked, thereby stopping transverse and horizontal movement of the scanning assembly **108**.

[0077] The module receiver **230** is positioned within the housing **310**. Specifically, the module receiver **230** is mechanically connected to a first end of the housing **310** at a rear side wall **328** of the frame **322**, and the first end is closer to the adjustable arm **106** than a second end of the housing **310**. The second end of the housing **310** is located at a front side wall **326** of the frame **322**. In one example, the module receiver **230** is connected to the first end via a protruding portion of the module receiver **230**, and the protruding portion is connected to a motor of the module receiver **230**.

[0078] As described above, the housing **310** is configured to remain stationary during scanning. In other words, once the weight applied to the scanning assembly **108** is adjusted by means of the adjustable arm **106** and then the ball joint **112** is locked, the housing **310** may remain in the resting position without translating in the horizontal or transverse direction. However, the housing **310** may still translate vertically as the adjustable arm **106** move vertically.

[0079] Instead, the module receiver **230** is configured to translate relative to the housing **310** during scanning. As shown in FIG. 3, the module receiver **230** translates horizontally along a horizontal axis **306** relative to the housing **310**. The motor of the module receiver **230** may slide the module receiver **230** along an upper surface of the first end of the housing **310**.

[0080] The transducer module **220** is removably connected to the module receiver **230**. Therefore, during scanning, the transducer module **220** and the module receiver **230** translate horizontally. During scanning, the transducer module **220** sweeps horizontally across the breast under the control of the motor of the module receiver **230**, and at the same time, a contact surface of the transducer module **220** contacts the film **118**. The transducer module **220** and the module receiver **230** are connected together at a module interface **320**. The module receiver **230** has a width that is the same as that of the transducer module **220**. In an alternative embodiment, the width of the module receiver may be different from that of the transducer module **220**. In some embodiments, the module interface **320** includes a connector between the transducer module **220** and the module receiver **230**, and the connector includes mechanical and electrical connections.

[0081] The foregoing provides an illustrative description of the structure of the scanning assembly **108**, but the present application is not limited thereto. The couplant coating mechanism according to the embodiments of the present application may alternatively be engaged with a scanning probe of a scanning assembly having another structure.

[0082] FIG. 6 is a schematic exploded view of a couplant coating mechanism according to embodiments of the present application.

[0083] As shown in FIG. 6, in some embodiments, an accommodation portion **420** further includes

an extension portion **422** connected to a main body portion **421**, the extension portion **422** is provided with a second channel **423** that is in communication with an internal space of the main body portion **421**, and the internal space of the main body portion **421** is in communication with the first channel **411** via the second channel **423**. Thus, a couplant accommodated in the main body portion **421** can flow into the first channel **411** along the second channel **423**, and flow out.

[0084] In some embodiments, as shown in FIG. 6, there are two extension portions **422** oppositely disposed, and each extension portion **422** is provided with the second channel **423** described above. Thus, the couplant accommodated in the main body portion **421** of the accommodation portion **420** can flow out from both sides in a moving direction of the scanning probe P, thereby improving uniformity and stability of couplant coating.

[0085] In some embodiments, as shown in FIG. 6, an engagement portion **410** has two wall portions **410a** and **410b**, a gap between the two wall portions **410a** and **410b** is used for accommodating the scanning probe P, that is, after the couplant coating mechanism is assembled to the scanning probe P, the two wall portions **410a** and **410b** are located on both sides in the moving direction of the scanning probe P, respectively, as shown in FIG. 4. In FIG. 4, the left-right direction is the moving direction of the scanning probe P.

[0086] In some embodiments, as shown in FIG. 6, there are a plurality of first channels **411** disposed extending from top to bottom; eight first channels are shown in FIG. 6. By means of providing a plurality of first channels **411**, uniformity of extrusion of the couplant can be ensured and usage can be reduced as much as possible.

[0087] In the foregoing embodiments, the quantity of the second channels **423** disposed on each extension portion **422** is the same as the quantity of the first channels **411** disposed on the wall portion of a corresponding side. As shown in FIG. 6, the quantity of the second channels **423** is the same as the quantity of the first channels **411**, and eight of both being provided. The quantities of the second channels **423** and the first channels **411** disposed on the extension portions **422** and the wall portions on different sides may be the same or different. For example, there are eight second channels **423** and first channels **411** on one side, and there are seven second channels **423** and first channels **411** on the other side, which may be specifically configured according to requirements.

[0088] In the foregoing embodiment, as shown in FIG. 6, the width of the first channel **411** may be larger than the width of the second channel **423**, and the projection of the second channel **423** in the up-down direction falls within the range of the first channel **411**. Thus, when the couplant flows out from the main body portion **421** of the accommodation portion **420**, the couplant may flow out first along the second channel **423** having a smaller width and then flow out along the first channel **411** having a larger width, thereby ensuring smooth flowing out of the couplant and avoiding clogging during the flowing of the couplant.

[0089] The foregoing is just an example, and the present application is not limited thereto. The first channel **411** and the second channel **423** may also have the same width and completely penetrate through one another, which can also ensure smooth flowing of the couplant and avoid clogging.

[0090] In the foregoing embodiment, the connection manner between the extension portion **422** and the engagement portion **410** is not limited as long as it is ensured that the second channel **423** and the first channel **411** can penetrate through, and the couplant can enter the first channel **411** from the second channel **423** and flow out from a couplant outlet **412** along the first channel **411**.

[0091] In some embodiments, as shown in FIG. 6, extending ends of the two wall portions **410a** and **410b** each have a couplant outlet **412** that is in communication with the first channel **411**. The through-type couplant outlet **412** can ensure that a small amount of couplant can be uniformly dispersed after passing through the first channel **411**, thereby achieving a uniform coating effect even in a case of a small amount of couplant.

[0092] In the foregoing embodiments, as shown in FIG. 6, the couplant outlet **412** may be an integral structure formed along the extending ends of the foregoing wall portions **410a** and **410b**, that is, the couplant outlet **412** penetrates through, and the couplant flowing out from any first

channel **411** flows into the couplant outlet **412**, and is coated on a skin surface of a patient from the couplant outlet **412**. For example, when the scanning probe P moves in one direction (for example, the right side in FIG. 4), the couplant flows out from the couplant outlet **412** on the right side in FIG. 4. When the scanning probe P moves in the other direction (for example, the left side in FIG. 4), the couplant flows out from the couplant outlet **412** on the left side in FIG. 4.

[0093] In some embodiments, as shown in FIG. 5, the curvature of the couplant outlet **412** matches that of an extending end of the scanning probe P. Thus, it can be ensured that the couplant outlet **412** does not affect the scanning of the patient during a scanning process.

[0094] In some embodiments, as shown in FIG. 5, the height of the couplant outlet **412** is higher than that of the extending end of the scanning probe P. Thus, it can be further ensured that the couplant outlet **412** does not affect the scanning of the patient during a scanning process.

[0095] FIG. 7 is a schematic diagram of a partial structure of an accommodation portion **420** of a couplant coating mechanism according to embodiments of the present application.

[0096] As shown in FIG. 6 and FIG. 7, in some embodiments, the accommodation portion **420** further includes an extruding portion **424** disposed inside the main body portion **421** and a driving portion **425** that drives the extruding portion **424** to move. The extruding portion **424** moves under the driving of the driving portion **425** to extrude the couplant inside the main body portion **421**. Thus, the couplant accommodated in the main body portion **421** can flow into the couplant outlet **412** along the second channel **423** and the first channel **411**, thereby implementing automatic coating of the couplant.

[0097] In some embodiments, as shown in FIG. 6 and FIG. 7, the extruding portion **424** includes a first flipping plate **424a** and a second flipping plate **424b** that divide the internal space of the main body portion **421** into two parts, and the driving portion **425** includes a first motor **425a** and a second motor **425b**. The first motor **425a** is capable of driving the first flipping plate **424a** to move, so that the couplant on a side of the first flipping plate **424a** flows out; and the second motor **425b** is capable of driving the second flipping plate **424b** to move, so that the couplant on a side of the second flipping plate **424b** flows out.

[0098] FIG. 8 is another schematic diagram of a couplant coating mechanism according to embodiments of the present application, showing a case in which the couplant coating mechanism is viewed from a side where a motor is located (for example, the left side shown in FIG. 6 or the right side shown in FIG. 6). FIG. 9 is yet another schematic diagram of a couplant coating mechanism according to embodiments of the present application, showing a case in which the couplant coating mechanism is viewed from a side where the wall portion **410a** or **410b** of the engagement portion **410** is located (for example, the left side shown in FIG. 4 or the right side shown in FIG. 4).

[0099] As shown in FIG. 8 and FIG. 9, when a scanning probe moves in a first direction, for example, when the scanning probe P moves to the left side shown in FIG. 4, a first motor **425a** drives a first flipping plate **424a** to rotate, and a couplant flows downward along a second channel **423** and a first channel **411** under the pressure of the first flipping plate **424a**, then flows out from a couplant outlet **412**, and is applied to a skin surface of a patient. Similarly, when the scanning probe moves in a second direction, for example, when the scanning probe P moves to the right side shown in FIG. 4, a second motor **425b** drives a second flipping plate **424b** to rotate, and the couplant flows downward along the second channel **423** and the first channel **411** under the pressure of the second flipping plate **424b**, then flows out from the couplant outlet **412**, and is applied to the skin surface of the patient. Thus, by means of the couplant coating mechanism, automatic application of a couplant is implemented, thereby saving scanning time, avoiding couplant waste, and further improving scanning efficiency and user experience.

[0100] According to the foregoing embodiment, in some possible implementations, the driving of the extruding portion **424** by the driving portion **425** is earlier than the movement of the scanning probe P. For example, as shown in FIG. 4, before the scanning probe P moves to the left side, the

first motor **425a** operates before the scanning probe P moves. After the couplant flows out from the couplant outlet **412** on the left side, the scanning probe P moves to the left side and starts operating. When the scanning probe P reaches the leftmost side, the first motor **425a** stops operating, and the second motor **425b** starts operating. After the couplant flows out from the couplant outlet **412** on the right side, the scanning probe P moves to the right side and starts operating. Thus, automatic coating of the couplant during a scanning process is implemented. The foregoing is just an example, and during a specific implementation process, the driving portion **425** and the scanning probe P may also operate at the same time, specifically depending on requirements.

[0101] In some embodiments, as shown in FIG. 4 to FIG. 6, the main body portion **421** is provided with two openings **426**, and the two openings **426** are in communication with the two internal spaces divided by the first flipping plate **424a** and the second flipping plate **424b**, respectively, and are used by an operator to inject a couplant into the main body portion **421**. Configured locations, sizes, and shapes of the openings **426** are not limited in the present application, depending on a specific requirement.

[0102] In some embodiments, the main body portion **421** further includes an electrical connection portion (not shown in the figures). The electrical connection portion is connected to the driving portion **425** to transmit electrical energy to the driving portion **425**.

[0103] In the foregoing embodiments, a configured location of the electrical connection portion is not limited. In one possible implementation, the electrical connection portion includes an electrical receiving terminal. The electrical receiving terminal may be disposed on an inner side of the main body portion **421** or the engagement portion **410** and cooperates with a power supply terminal of the scanning probe P to receive the foregoing electrical energy. In another possible implementation, the electrical connection portion includes a cable. The cable receives electrical energy from outside the couplant coating mechanism.

[0104] The above embodiments merely provide illustrative descriptions of the embodiments of the present application. However, the present application is not limited thereto, and appropriate variations may be made on the basis of the above embodiments. For example, each of the above embodiments may be used independently, or one or more among the above embodiments may be combined.

[0105] In addition, the foregoing merely describes the structure of the couplant coating mechanism related to invention points of the present application. In a specific implementation, the couplant coating mechanism may further include other components used to implement other functions. As shown in FIG. 6, the accommodation portion **420** of the couplant coating mechanism may further include cover portions **427** between the motors **425a/425b** and the main body portion **421** for covering both ends of the main body portion **421**, a sealing member **428** for sealing the openings **426**, and the like.

[0106] It can be learned from the foregoing embodiments that automatic application of a couplant is implemented by adding the couplant coating mechanism to the scanning probe, thereby saving scanning time, avoiding couplant waste, and further improving scanning efficiency and user experience. In addition, by means of the above structure, an operator only needs to install the couplant coating mechanism on the scanning probe to implement automatic coating of a couplant, thereby improving the convenience of coating. In addition, since the couplant coating mechanism is engaged with the scanning probe, the couplant coating mechanism can be easily removed from the scanning probe when the couplant coating mechanism is not required, thereby improving convenience of use.

[0107] The embodiments of the present application further provide an ultrasonic imaging system. The ultrasonic imaging system includes a scanning assembly and the couplant coating mechanism described in the foregoing embodiments. Related content of the couplant coating mechanism has been described above, which will not be described again here.

[0108] In the foregoing embodiments, the scanning assembly may further include a frame, for

example, the frame 322 shown in FIG. 3. The frame 322 accommodates the described scanning probe (the transducer module 220 and the module receiver 230) and the driving device 240. The driving device 240 drives the scanning probe to move within the frame 322 to perform ultrasonic scanning. Since the composition and the operating manner of the scanning assembly are already described in the foregoing embodiments, details are not described again here.

[0109] In the embodiments of the present application, the combination of an automatically-driven scanning assembly and a couplant coating mechanism can improve scanning efficiency while improving user experience, especially as compared with a conventional hand-held ultrasonic probe. In one aspect, opposing upper surfaces of the scanning probe and the transducer in the scanning assembly provide sites of engagement for the couplant coating mechanism, whereas opposing sides of a transducer of a hand-held probe typically have cables or parts that need to be grasped by a physician, which makes it difficult to secure a couplant application mechanism. In another aspect, since the scanning probe in the scanning assembly is driven by the driving device, an additional couplant coating mechanism does not cause the physician to feel a lowering of user experience such as an increase in weight and an increase in volume, whereas it is difficult for a hand-held probe to achieve the described effect.

[0110] The present application is described above with reference to specific embodiments. However, it should be clear to those skilled in the art that the foregoing description is merely illustrative and is not intended to limit the scope of protection of the present application. Various variations and modifications may be made by those skilled in the art according to the spirit and principle of the present application, and these variations and modifications also fall within the scope of the present application.

[0111] Preferred embodiments of the present application are described above with reference to the accompanying drawings. Many features and advantages of the implementations are clear according to the detailed description, and therefore the appended claims are intended to cover all these features and advantages that fall within the true spirit and scope of these implementations. In addition, as many modifications and changes could be easily conceived of by those skilled in the art, the embodiments of the present application are not limited to the illustrated and described precise structures and operations, but can encompass all appropriate modifications, changes, and equivalents that fall within the scope of the implementations.

Claims

1. A couplant coating mechanism, comprising: an engagement portion engaging with a scanning probe, the engagement portion being provided with a first channel; and an accommodation portion connected to the engagement portion, the accommodation portion comprising a main body portion for accommodating a couplant, an internal space of the main body portion being in communication with the first channel, and the couplant accommodated in the main body portion flowing out along the first channel.
2. The couplant coating mechanism according to claim 1, wherein the accommodation portion further comprises an extension portion connected to the main body portion, the extension portion is provided with a second channel that is in communication with the internal space of the main body portion, and the internal space of the main body portion is in communication with the first channel via the second channel.
3. The couplant coating mechanism according to claim 2, wherein there are two extension portions oppositely disposed, and each extension portion is provided with the second channel.
4. The couplant coating mechanism according to claim 1, wherein the engagement portion comprises two wall portions, and a connection portion connecting the two wall portions; and a gap between the two wall portions is used for accommodating the scanning probe, and each wall portion is provided with the first channel.

5. The couplant coating mechanism according to claim 4, wherein there are a plurality of first channels disposed extending from top to bottom.
6. The couplant coating mechanism according to claim 4, wherein extending ends of the two wall portions each have a couplant outlet in communication with the first channel.
7. The couplant coating mechanism according to claim 6, wherein the couplant outlet is an integral structure formed along the extending end, and the curvature of the couplant outlet matches that of an extending end of the scanning probe.
8. The couplant coating mechanism according to claim 6, wherein the height of the couplant outlet is higher than the height of an extending end of the scanning probe.
9. The couplant coating mechanism according to claim 1, wherein the accommodation portion further comprises: an extruding portion, the extruding portion being disposed inside the main body portion and being capable of moving to extrude the couplant; and a driving portion, the driving portion driving the extruding portion to move.
10. The couplant coating mechanism according to claim 9, wherein the extruding portion comprises a first flipping plate and a second flipping plate that divide the internal space of the main body portion into two parts, and the driving portion comprises a first motor and a second motor; and the first motor is configured to drive the first flipping plate to move, so that the couplant on a side of the first flipping plate flows out, and the second motor is configured to drive the second flipping plate to move, so that the couplant on a side of the second flipping plate flows out.
11. The couplant coating mechanism according to claim 10, wherein the main body portion is provided with two openings, the two openings being in communication with two internal spaces, respectively, which are separated by the first flipping plate and the second flipping plate.
12. The couplant coating mechanism according to claim 9, wherein the main body portion further comprises: an electrical connection portion, the electrical connection portion being connected to the driving portion to transmit electrical energy to the driving portion, and the electrical connection portion comprising at least one of: an electrical receiving terminal, the electrical receiving terminal being disposed on an inner side of the main body portion or the engagement portion, and cooperating with a power supply terminal of the scanning probe to receive the electrical energy; and a cable, the cable receiving electrical energy from outside the couplant coating mechanism.
13. An ultrasonic imaging system, characterized by comprising: a scanning assembly, comprising a scanning probe; and the couplant coating mechanism comprising: an engagement portion engaging with a scanning probe, the engagement portion being provided with a first channel; and an accommodation portion connected to the engagement portion, the accommodation portion comprising a main body portion for accommodating a couplant, an internal space of the main body portion being in communication with the first channel, and the couplant accommodated in the main body portion flowing out along the first channel.
14. The ultrasonic imaging system according to claim 13, wherein the scanning assembly further comprises a frame, the frame accommodating the scanning probe and a driving device, the driving device driving the scanning probe to move within the frame to perform ultrasonic scanning.
15. The couplant coating mechanism according to claim 13, wherein the accommodation portion further comprises an extension portion connected to the main body portion, the extension portion is provided with a second channel that is in communication with the internal space of the main body portion, and the internal space of the main body portion is in communication with the first channel via the second channel.
16. The couplant coating mechanism according to claim 15, wherein there are two extension portions oppositely disposed, and each extension portion is provided with the second channel.
17. The couplant coating mechanism according to claim 13, wherein the engagement portion comprises two wall portions, and a connection portion connecting the two wall portions; and a gap between the two wall portions is used for accommodating the scanning probe, and each wall portion is provided with the first channel.

18. The couplant coating mechanism according to claim 17, wherein there are a plurality of first channels disposed extending from top to bottom.

19. The couplant coating mechanism according to claim 17, wherein extending ends of the two wall portions each have a couplant outlet in communication with the first channel.

20. The couplant coating mechanism according to claim 19, wherein the couplant outlet is an integral structure formed along the extending end, and the curvature of the couplant outlet matches that of an extending end of the scanning probe.
